Saidakhmad Nuriddinov

Autonomous Driving Simulation Engineer

CONTACT

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SKILLS

Robotics & Simulation
NVIDIA Isaac Sim, Omniverse, USD, PhysX, OmniGraph, ROS 2, Nucleus

Languages
Python, C++, TypeScript, JavaScript

Frontend
React, Next.js, SCSS

Backend
NestJS, Node.js, Flask

Database & DevOps
PostgreSQL, MySQL, MariaDB, Docker, Vercel, GitHub Pages

Tools
Git, Postman, Notion, Logitech G29

LANGUAGES

Korean — Advanced · TOPIK 4

English — Advanced · IELTS 7

CERTIFICATES

KIIP Level 5
Nov 2025

Wcoding Full-Stack Bootcamp
Nov 2024

TOPIK Level 4
Jul 2023

Business Korean, Seoul Global Center
Sep 2023

Minister's Prize, Education Ministry of Uganda
Jun 2023

President's Award, KIU Uganda
Jun 2023

Dean's Award, Computer Engineering
Dec 2021

EDUCATION

Gachon University
Seongnam, South Korea · Aug 2018 – Feb 2023

B.S. in Computer Engineering

EXPERIENCE

Maum.ai (마음에이아이)
Seongnam · Feb 2026 – May 2026

Autonomous Driving Simulation Engineer · WoRV

Autonomous yard-tractor simulation (`ability_port`, XCMG 15-DOF) on NVIDIA Isaac Sim / Omniverse.

- Built worv.env.lights, a solar lighting Isaac Sim extension using a stdlib-only Fourier model; established the MVC pattern (extension → window → controller) adopted as the team-wide template for all worv.* extensions.
- Shipped worv.env.climate (rain/snow/fog via USD PointInstancer + Warp + RTX) and led a 5-bug fix branch introducing a 3-source PositionTracker for robot-following rain and controller-level error surfacing.
- Designed worv.core.logging, a configurable-Hz simulation logger combining OmniGraph with direct PhysX USD reads via a pluggable BaseExporter — enabled the team's first benchmark scenario comparison.
- Built a Docker Nucleus worker plus a 3-commit UE5 C++ plugin patch fixing UE Connector's baked 1×1 opacity textures so designers can edit opacity directly in Isaac Sim.
- `ability_port` — Applied physics, colliders, and NavMesh across the map; added an Ackermann controller and refactored the ROS 2 `teleop_node` from Xbox-only to hardware-agnostic via a JoyMapping dataclass.

Stack: NVIDIA Isaac Sim · Omniverse · USD · PhysX · OmniGraph · Nucleus · ROS 2 · Python · C++ (UE5) · Docker

EARLIER EXPERIENCE

ChoiceTech Korea (초이스테크코리아)
Yongin · Jul 2025 – Oct 2025

Backend Developer

AI-powered hair & skin analysis platforms used by L'Oréal and Dior.

- Implemented JWT auth, CRUD, and pagination on NestJS + PostgreSQL; shipped new features and bug fixes that improved backend stability and maintainability.
- Customized backend APIs and database structures for L'Oréal and Dior analysis platforms; implemented real-time update mechanisms tailored to each brand.
- Validated web/app analysis-result consistency on Next.js, integrated REST APIs, and resolved cross-stack data-flow issues end-to-end.
- Standardized license-expiry logic across all projects with automated daily updates and 30-day pre-expiration notifications.
- Optimized SQL keyword mapping for dermatological analysis, achieving over 20% faster real-time response.

Stack: NestJS · PostgreSQL · Next.js · React · TypeScript · JavaScript

STEMON (스템온)

Seongnam · Dec 2024 – Jul 2025

Software Developer

Web services and embedded medical-device management systems.

- Built a corporate Q&A system (HTML/CSS/JS) and an investor-relations React/SCSS SPA; deployed via GitHub Pages and Vercel with CI/CD.
- Developed the DDDS Admin App in C++ for real-time medical-device monitoring with auto-detection of faulty components and CSV export.
- Validated hardware-software integration with ESP32-S3 over USB serial; redesigned the on-device UI on DWIN DGUS for production deployments.
- Led the final-month team project as Wcoding bootcamp lead — full-stack React + Flask + MySQL with auth, notifications, and DB design.

Stack: React · SCSS · JavaScript · HTML/CSS · Flask · MySQL · C++

SELECTED PROJECTS

ParkPilot Personal · 2026.06

RL auto-parking agent for Ackermann (car-like) vehicles. MuJoCo + SAC, CPU-only on Apple Silicon — no NVIDIA, no Isaac Sim. 99.5% success on randomized starts, 100% on fixed starts across 2,835 evaluation episodes. `github.com/Saidakhmad77/parkpilot`

worv.core.logging Maum.ai · 2026.04

Configurable-Hz simulation data logger. Hybrid OmniGraph + direct PhysX USD reads. Pluggable BaseExporter (CSV/JSON/Console), smart root-link resolution, frame dedup via `absoluteSimTime`.

G29 Teleop Port Maum.ai · 2026.03 – 2026.04

Hardware-agnostic ROS 2 teleop for the XCMG tractor. Refactored hardcoded Xbox enum into a configurable JoyMapping dataclass with per-axis inversion, deadzone gating, accel/brake pedal merging.

DDDS Admin App STEMON · 2025.03 – 2025.07

Internal real-time device monitoring (C++) with ESP32-S3 USB serial integration and DWIN DGUS embedded UI; auto-detection of faulty parts and CSV data export.

worv.env.climate Maum.ai · 2026.04

Rain/snow/fog effects via USD PointInstancer + Warp kernels + RTX carb settings. 5-bug fix branch (+405/−358) including PositionTracker, RTX key discovery, silent-failure surfacing pattern.

usd-opacity-tools Pipeline Maum.ai · 2026.04

Docker Nucleus post-processor + 3-commit UE5 C++ plugin patch fixing UE Connector's baked 1×1 opacity textures so designers can edit opacity in Isaac Sim. Includes DeviceFlow auth workaround for headless containers.

worv.env.lights Maum.ai · 2026.03

Solar lighting Isaac Sim extension. Spencer 1971 Fourier solar model (stdlib only), Tanner Helland Kelvin→RGB, timeline 1×–500×, adopt-vs-create lights, stage lifecycle subscription. MVC architecture became the team-wide template.

EXTRACURRICULAR

Full-Stack Bootcamp, Wcoding
Team Lead · Jul – Nov 2024

Leaders Conference, IYF
Team Leader (Uganda) · 2023

Business Korean Program
Seoul Global Center · Jul – Sep 2023

Teaching Assistant
Gachon University · Mar – Jun 2022